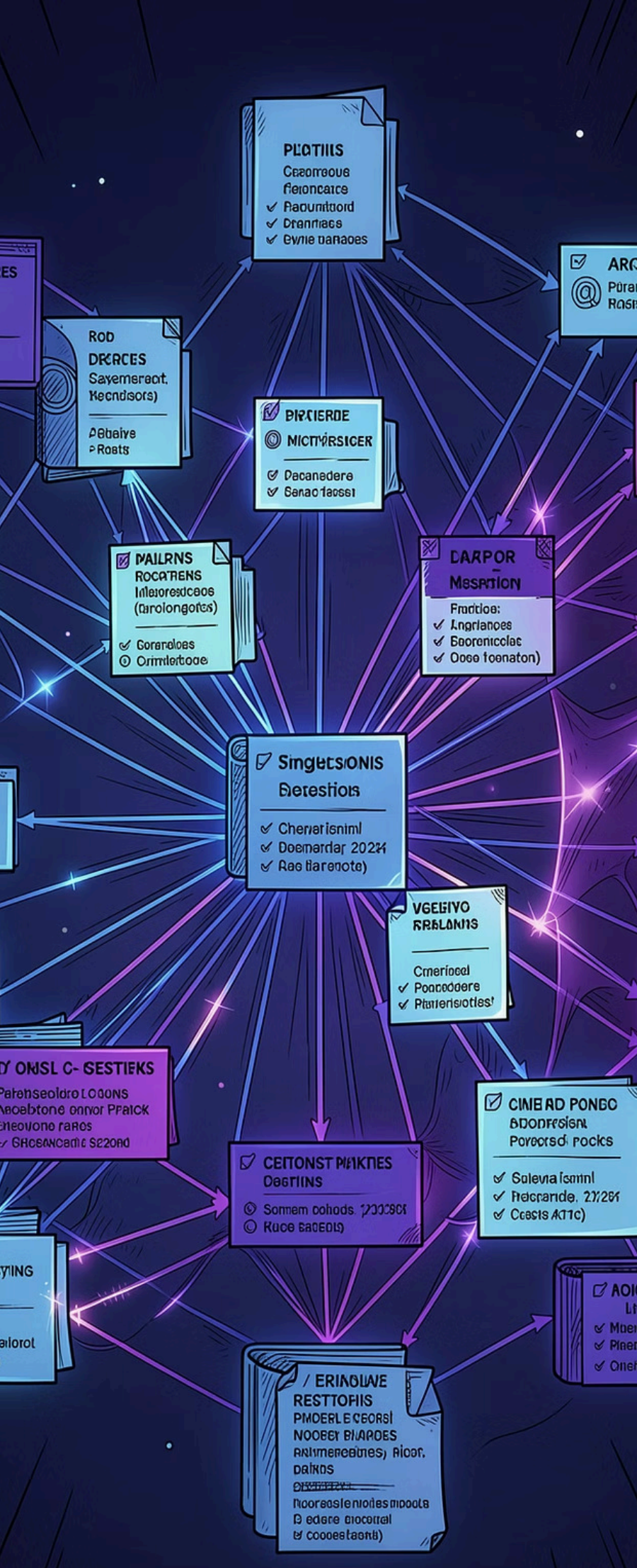




CiteGraph-NLP

Confidence-Aware Citation Lineage and Study-Scale Knowledge Graph System

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Project Scope and Requirements

Scope

- Input a seed paper (DOI, PMID, PMCID, title, URL)
- Build citation lineage (backward + forward traversal)
- Extract population evidence (N_eff) from paper text
- Rank probable foundational papers using weighted graph analytics

Requirements

- Multi-provider metadata coverage (OpenAlex, Crossref, Europe PMC)
- Confidence-aware extraction and ambiguity handling
- Usable dashboard for metadata, graph, paths, rankings, and export
- Deployable stack: frontend + backend with reproducible setup

Pipeline Overview

End-to-End Pipeline

01	02	03
Input Normalization	Metadata Resolution	Citation Traversal
	Parallel providers	BFS, depth-limited
04	05	06
Population Extraction	Population Resolution	Edge Weighting & Graph Analytics
Regex + semantic typing	n_eff selection	
07		
Output / Export + Visualization		

Core Output

- Probable foundational papers
- Ranked citation paths
- Confidence-aware population evidence

Pipeline Details

Key Processing Logic

Metadata Merge

Field-level precedence across providers.

Traversal Control

Configurable depth and max paper cap.

Population Resolver

Picks best candidate using confidence + semantic priority.

Weighting Formula

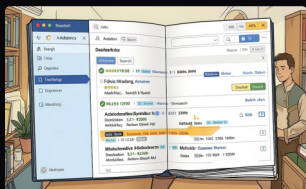
```
N_score = log1p(N_eff) / log1p(N_reference)
base_weight = alpha*N_score + beta*journal_score
final_weight = base_weight * pop_confidence * edge_confidence
```

Foundational Ranking

```
Combined score:
0.5*PageRank + 0.3*year_score + 0.2*evidence_score
```

Dataset(s)

Data Sources



OpenAlex

Metadata, references, forward citations



Crossref

DOI metadata, title search, references



Europe PMC

Biomedical metadata and ID resolution

Input Dataset Nature

- Dynamic retrieval from public scholarly APIs (no fixed static dataset)
- Supports biomedical and clinical literature best
- Optional local PDF parsing path for richer extraction

Models and Methods

Methods Used

NLP Extraction

Pattern-based numeric extraction with semantic classes

Confidence Scoring

Rule-based confidence + ambiguity status

Graph Modeling

NetworkX directed weighted graph

Ranking Model

Weighted PageRank + temporal + evidence signals

Why This Choice

- Interpretable outputs
- Fast iteration for research prototype
- Confidence visibility over black-box scoring

Evaluations

Technical Evaluation

Unit and integration tests for:

- Input validation and normalization
- API run lifecycle
- Citation retrieval / traversal
- Ranking stability and export endpoints

Functional Evaluation (Example Run)

End-to-end run returns structured outputs:

- Papers
- Population resolutions
- Citation edges
- Ranked paths
- Foundational rankings

Quality Signals Tracked

Extraction Status

Resolved, ambiguous, or missing

Confidence Scores

Per candidate and resolution

Graph / Ranking Consistency

Under fixed settings

Strengths and Limitations

Strengths

- Confidence-aware design (not just raw extraction)
- Multi-source citation + metadata integration
- Explainable ranking logic (formula-based)
- Full-stack implementation with deploy pipeline

Limitations

- Population extraction is regex-based; misses complex phrasing
- Many papers have sparse abstracts, reducing extractable evidence
- Citation coverage depends on external provider completeness
- Rankings are probabilistic, not absolute ground truth

Future Direction

Next Improvements

→ Stronger Extraction Model

Add hybrid rules + ML/LLM-based information extraction

→ Full-Text Parsing Pipeline

Expand to Methods sections for richer evidence

→ Better Disambiguation

Improve trial-level entity resolution across papers

→ Benchmark Suite

Compare against tools like Semantic Scholar / ResearchRabbit

→ Robustness

Improve resilience under provider rate limits and network instability

Product Direction

Collaboration & Export

Better collaboration and export features for research teams

Explainability Overlays

Enhanced explainability in graph and path views

Running Demo with Code

Local Demo Steps

```
cd <parent_dirs>\CiteGraph-NLP
.\start-local.ps1 -Install
```

Manual Start (Alternative)

```
# Backend
.\.venv\Scripts\Activate.ps1
$env:PYTHONPATH="$PWD\src"
uvicorn citegraph.api.main:app --reload --host 127.0.0.1
--port 8000

# Frontend (new terminal)
cd frontend
npm install
npm run dev
```

Demo URL + Sample Input

- **Frontend:** <http://localhost:5173> or <https://citegraph-nlp1.web.app/>
- **API:** <http://localhost:8000/docs>
- **Health Check:** <https://hetzner-api.duckdns.org/citegraph/health>
- **Sample DOI:** 10.1038/s41467-021-23458-5
- **Paper Used in Demo:** [Nature Communications](#)
- **Result Snapshot:** [CiteGraph-NLP — Research Graph Intelligence](#)